NLP-Based Customer Support Ticket Classification System

Milestone 1: Project Proposal

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This project aims to develop an NLP-based customer support ticket classification system for logistics and e-commerce companies. The system will analyze customer messages and automatically classify them into categories such as delivery issues, refund requests, damaged products, missing packages, and other service-related issues.

Machine learning and NLP techniques will be used to process and understand customer text data, helping companies improve customer support efficiency, and reduce manual workload.

Problem Statement

Logistics and e-commerce companies receive thousands of customer support requests every day related to delivery delays, damaged products, refund requests, missing packages, and other service-related issues. Managing and organizing these customer support tickets manually can be time-consuming and inefficient, especially as the number of customer requests continues to increase.

Companies require faster and more efficient methods for handling customer complaints while maintaining high-quality customer service. This project will use Natural Language Processing (NLP) and machine learning techniques to automatically classify customer support tickets based on the issue described in each message. The system aims to improve customer support efficiency, reduce manual workload, and decrease the time required to manage and organize support requests.

Data Selection

This project will utilize the *Customer Support Tickets Dataset* available through Hugging Face. The dataset contains customer support ticket messages related to logistics and e-commerce services and is appropriate for Natural Language Processing (NLP) and machine learning classification tasks.

The dataset consists of structured customer support records that include text-based customer complaints and service requests involving topics such as delivery delays, refund requests, damaged products, missing packages, and other customer support issues. These text records provide suitable data for preprocessing, feature extractions, and other customer support issues. These text records provide suitable data for preprocessing, feature extractions, and supervised machine learning model development.

This dataset was selected because it contains real-world customer support interactions that can be used to train and evaluate a classification system capable of automatically categorizing support tickets based on the issue described within the message. Additionally, the dataset supports the development of an automated ticket classification system capable of improving customer service efficiency and reducing manual workload.

The dataset also contains a large collection of customer support tickets that provide sufficient data volume for training, validation, and testing machine learning models. The structured nature of the dataset also allows for easier implementation of preprocessing techniques such as text cleaning, tokenization, stop-word removal, and feature extraction methods including TF-IDF and vectorization.

Prior to model training, the dataset will undergo preprocessing procedures to improve text quality and prepare the data for machine learning analysis. These preprocessing steps will help improve classification accuracy and overall model performance.

Data Source:

Customer Support Tickets Dataset - Hugging Face

Column #	Column Name	Non-Null Count	Type
0	Customer Email	8469	String
1	Product Purchased	8469	String
2	Ticket Type	8469	String
3	Ticket Subject	8469	String
4	Combined Text	8469	String
5	Ticket Priority	8469	String

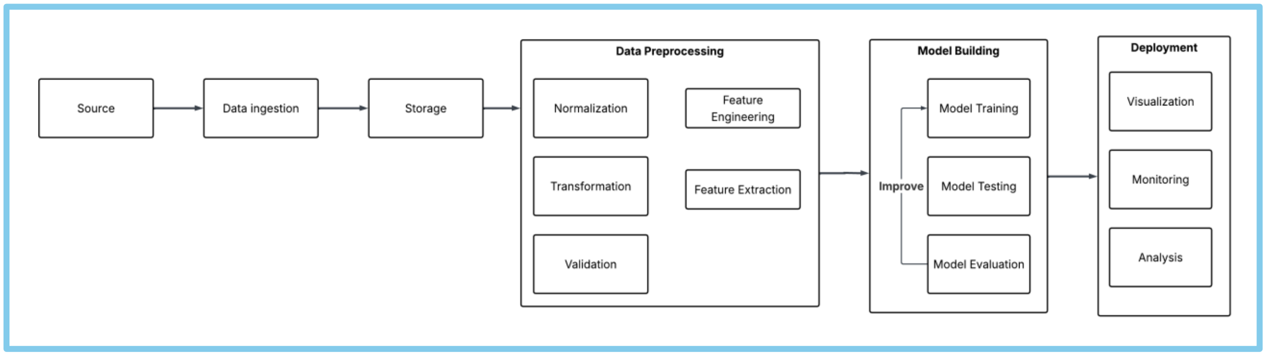
Dataset size: (8469, 6)df[tic

Expected Outcomes

At the completion of this project, we expect to have an NLP model that can successfully classify the ticket by type and subject and assign it a priority.

Methodology

This project will have the following pipeline:



Pipeline Stage	Tools and Methods	Objective
Data Collection	Hugging Face Dataset, CSV Files, Parquet files	Collect customer support messages
Data Preparation	Python, NLP	Clean and process text by removing unnecessary words and formatting the data
Feature Extraction	TF-IDF, Count Vectorizer	Convert into numerical features
Model Training	Scikit-learn, Naive Bayes	Train a machine learning model to classify customer support tickets into categories.
Model Testing	Accuracy Metrics	Test the model and evaluate its accuracy and performance.
Predication System	Machine Learning Model	Used trained model to automatically classify new customer support tickets.

Ticket Type: ['technical issue', 'billing inquiry', 'cancellation request', 'product inquiry', 'refund request']

Ticket Subject: ['product setup', 'peripheral compatibility', 'network problem', 'account access', 'data loss', 'payment issue', 'refund request', 'battery life', 'installation support', 'software bug', 'hardware issue', 'product recommendation', 'delivery problem', 'display issue', 'cancellation request', 'product compatibility']

Ticket Priority: ['critical', 'low', 'high', 'medium']

GitHub repository link: https://github.com/AlejandroTovarC/nlp-logistics-classification-system

Team Roles

This project will be completed by 3 students. All team members will collaborate equally throughout the project, including dataset preparation, text processing, model training, testing, evaluation, application deployment, and final documentation. Responsibilities will be among all members to support a fully collaborative approach to developing the NLP-based customer support ticket classification system.